

The Emacs Series

Exploring ts.el

In this article, the author reviews the ts.el date and time library for Emacs. Written by Adam Porter, ts.el is still in the development phase and has been released under the GNU General Public License v3.0.



The ts.el package uses intuitive names for date and time functions. It internally uses UNIX timestamps and depends on both the 'dash' and 's' Emacs libraries. The parts of the date are computed lazily and also cached for performance. The source code is available at <https://github.com/alphapapa/ts.el>. In this article, we will explore the API functions available from the ts.el library.

Installation

The package does not have a tagged release yet; hence, you should download it from <https://github.com/alphapapa/ts.el/blob/master/ts.el> and add it to your Emacs load path to use it. You should also have the 'dash' and 's' libraries installed and loaded in your Emacs environment. You can then load the library using the following command:

```
(require 'ts)
```

Usage

Let us explore the various functions available to retrieve parts of the date from the ts.el library. When the examples were executed, the date was 'Friday July 5, 2019'. The `ts-dow` function can be used to obtain the day of the week, as shown below:

```
(ts-dow (ts-now))
5
```

`ts-now` is a Lisp construct that returns a timestamp set. It

is defined in ts.el as follows:

```
(defsubst ts-now ()
  "Return 'ts' struct set to now."
  (make-ts :unix (float-time)))
```

The day of the week starts from Monday (1) and hence Friday has the value of 5. An abbreviated form of the day can be fetched using the `ts-day-abbr` function. In the following example, 'Friday' is shortened to 'Fri'.

```
(ts-day-abbr (ts-now))
"Fri"
```

The day of the week in full form can be obtained using the `ts-day-name` function, as shown below:

```
(ts-day-name (ts-now))
"Friday"
```

The twelve months from January to December are numbered from 1 to 12 respectively. Hence, for the month of July, the index number is 7. This numeric value for the month can be retrieved using the 'ts-month' API. For example:

```
(ts-month (ts-now))
7
```

If you want a three-character abbreviation for the